Samples and Chance — Wrong Denominators and Loaded Samples

Two questions catch most of the mistakes in this unit. Ask them every time.

- **"Out of what?"** A probability compares the outcomes you want with *every* outcome that could happen — not with the ones you do not want, and not with a smaller group inside the whole.
- **"Who got asked?"** A sample only speaks for a group if every member of that group had a real chance of being picked. Where a survey is taken can decide its answer before anyone replies.
- Give probabilities as fractions in lowest terms. Show the denominator you chose, and write one line saying what it counts.

Marble bag (problems 1 to 3): a bag holds 5 red, 3 blue and 4 green marbles, and one marble is drawn without looking.

1. **Warm-up.** Find $P(\text{red})$ for a single draw from the full bag.

Answer: __________________

2. **Find and fix.** Ines writes $P(\text{blue}) = \dfrac{3}{9}$, because there are 3 blue marbles and 9 marbles that are not blue.

Name what Ines put in the denominator, say what belongs there instead, and give the correct probability in lowest terms.

Answer: __________________

3. Find and fix. One red marble is drawn and *kept out of the bag.* Marco now says $P(\text{green}) = \dfrac{4}{12}$, "because there are still 4 green marbles."

Marco updated the numerator but not the denominator. What is the total number of marbles now, and what is the correct probability of drawing green?

Answer: ____________________

4. Find and fix. In a class of 24 students, 10 walk to school and 14 ride. Of the 10 walkers, 6 are in Year 7. One student is chosen at random from the class. Max writes $P(\text{a Year-7 walker}) = \dfrac{6}{10}$.

Max used the walkers as the whole group, but the student was chosen from the class. Give the correct probability in lowest terms.

Answer: ____________________

5. Find and fix. Nora wants to estimate how many of her school's 300 students cycle to school. She asks 20 students standing at the bike rack, and 18 of them say they cycle.

(a) If Nora's 20 students really did represent the school, how many of the 300 would cycle?
(b) Explain in one or two sentences why that number shows Nora's sample cannot be trusted, and name the feature of *where she asked* that caused it.

Answer: ________________

6. Find and fix. Sam surveys students in the school library about their favourite after-school club and records this:

Club	Chess	Art	Sport	Drama
Students	22	12	10	6

Sam concludes: "chess is the most popular club in the school."

(a) How many students answered the survey altogether? **(b)** Explain why the *place* Sam surveyed makes the conclusion unsafe, and say what he should do instead.

Answer: ________________

7. Two samples. Each list shows the weekly reading hours reported by 8 students.

Sample A, from the school book club: 14, 7, 6, 11, 7, 9, 8, 10.
Sample B, from 8 students drawn at random from the roll: 0, 8, 3, 6, 5, 1, 3, 2.

(a) Find the median of Sample A. **(b)** Find the median of Sample B. **(c)** The school wants one number for a typical student. Which median should it use, and why is the other one misleading?

Answer: ________________

8. Show what you know. The student council wants to know what fraction of the school's 300 students would use a later bus. Ben plans to ask the 30 students already waiting for the current late bus.

(a) Explain why Ben's sample is biased, and say whether his estimate will come out too high or too low.
(b) Describe a sampling plan that would not be biased, and say what makes it fair.
(c) The council instead asks 150 students chosen at random, and 45 say they would use the later bus. Write that probability as a fraction in lowest terms, then estimate how many of the 300 students would use it.